Supplementary Materials for "WRF Cold-Pool Sensitivities Relevant to Winter-Ozone Modelling in the Uinta Basin"

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These supplementary materials retain bounded secondary diagnostics from the 2 February 2013 Uinta Basin WRF case. They do not change the article's controlled driving-product result or the status of its failed gates.

Table S1: Controlled process-treatment effects on the frozen station/common-AGL pipeline. Values are means of five hourly spatial medians; signs are positive/negative hours with exact zeros omitted. Planetary boundary-layer height is scheme defined across the boundary-layer/surface-layer treatment.

Metric	Slope radiation (X1 − X0)		MYJ/Eta (X2 − X1)	
	Mean	Signs	Mean	Signs
T2 bias (°C)	+0.001	3/1	+0.37	5/0
10 m wind absolute error (m s^{-1})	+0.001	2/2	-0.145	0/5
Surface-pressure bias (Pa)	-0.12	0/4	+6.3	5/0
Stability $S_{50:500}$ (K)	+0.004	3/1	+0.11	3/2
Transport wind $U_{50:500}$ (m s^{-1})	+0.006	4/0	+0.098	5/0
PBL height (m)	+0.01	3/1	-9.2	2/3

Table S2: Static-terrain diagnostics retained after the failed default-minus-fine gate and the closed coarse-member lineage check.

Contrast	Terrain difference	2 m response	Status
Default minus fine	7.9 m RMS; -55 to +60 m range	Elevation-band means 0.07–0.27 K; terrain correlation below 0.3 at 14:00–16:00 UTC	Predeclared conjunction failed
Coarse brackets	91.9/93.1 m RMS; extrema to -462/+487 m	Slope-band means 1.2–1.5 K; terrain correlation 0.50–0.67	Descriptive only; no predeclared gate
Coarse-member lineage	As-run namelist matches the controlled family; terrain is materially smoother	Configuration caveat closed	Supports controlled-bracket status, not a terrain ranking

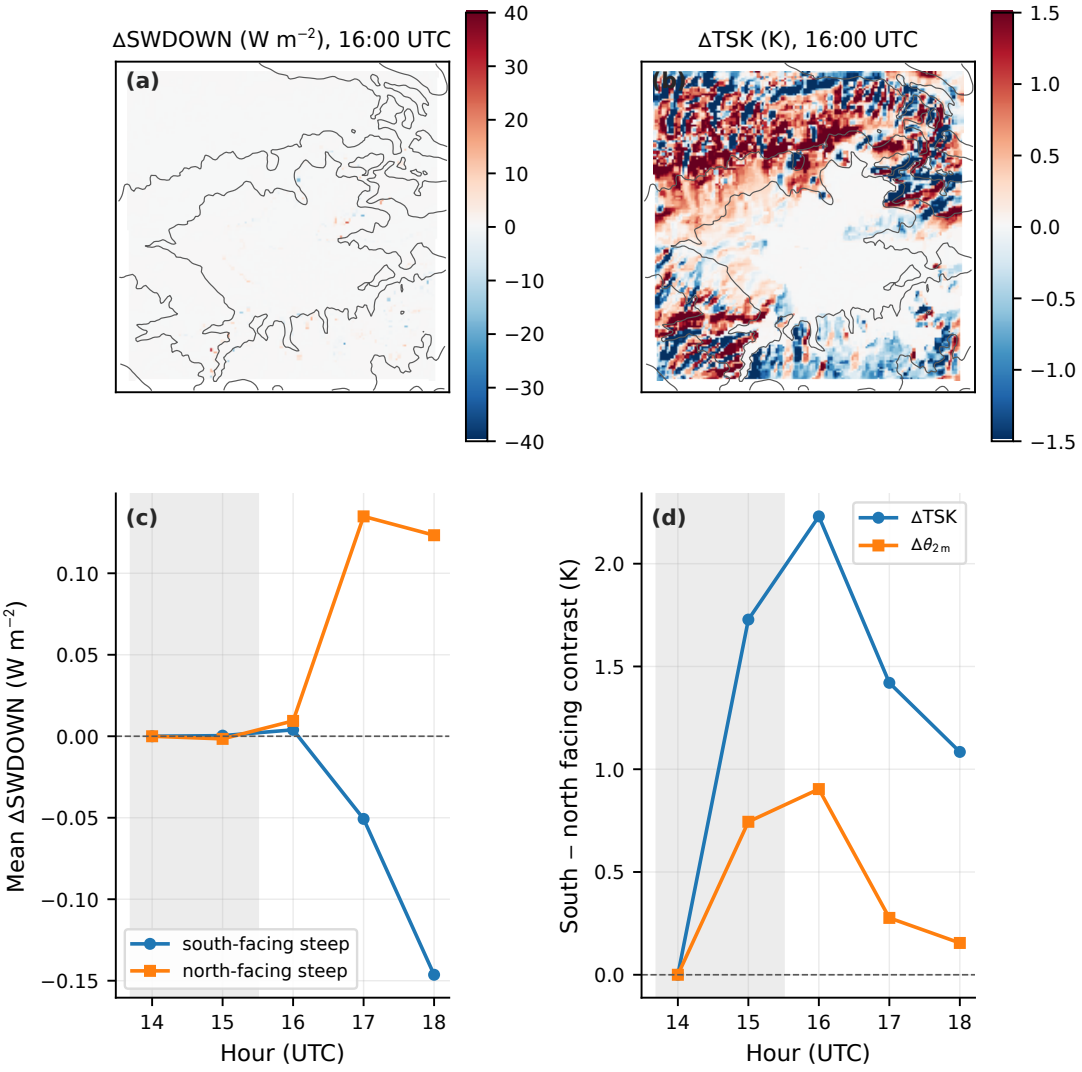


Figure S1: Diagnostic for the slope-radiation screen, calculated as X1 minus X0 on d02. The archived flat-plane shortwave field did not express the treatment, while skin temperature showed an aspect-organised response on steep terrain. The predeclared screen failed for this measurement-limited reason, and the figure does not support a configuration ranking or a claim that slope-aware radiation improved or degraded the cold pool.